



NORTH SYDNEY BOYS HIGH SCHOOL

PRELIMINARY MATHEMATICS

2015 Preliminary HSC Course Assessment Task 2

Thursday June 4, 2015

General instructions

- Working time – 50 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.
- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: # BOOKLETS USED:

Class (please ✓)

☐ 11M2A – Mr Zuber

Marker's use only.

QUESTION	1	2	3	4	Total	%
MARKS	$\overline{14}$	$\overline{8}$	$\overline{9}$	$\overline{14}$	$\overline{45}$	$\overline{100}$

45 marks

Attempt Questions 1 to 4

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 1 (14 Marks)

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Marks

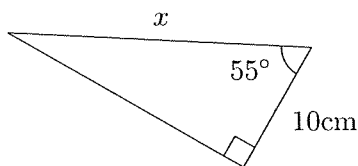
- (a) If $\sin \alpha = 0.72$ Find the value of the acute angle α , rounding correct to the nearest minute. **1**

- (b) Find the exact value of

i. $\cos 210^\circ$ **1**

ii. $\frac{2 \tan 60^\circ}{1 - \tan^2 60^\circ}$ **2**

- (c) Find the value of x , correct to 1 decimal place **2**

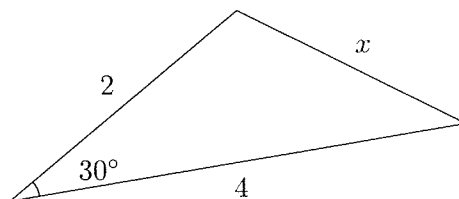


- (d) If $\cos \theta = -\frac{3}{7}$ for $0 \leq \theta \leq 360^\circ$ Find the value of

i. $\sec \theta$ **1**

ii. $\sin \theta$, given $\sin \theta$ is negative. **3**

- (e)



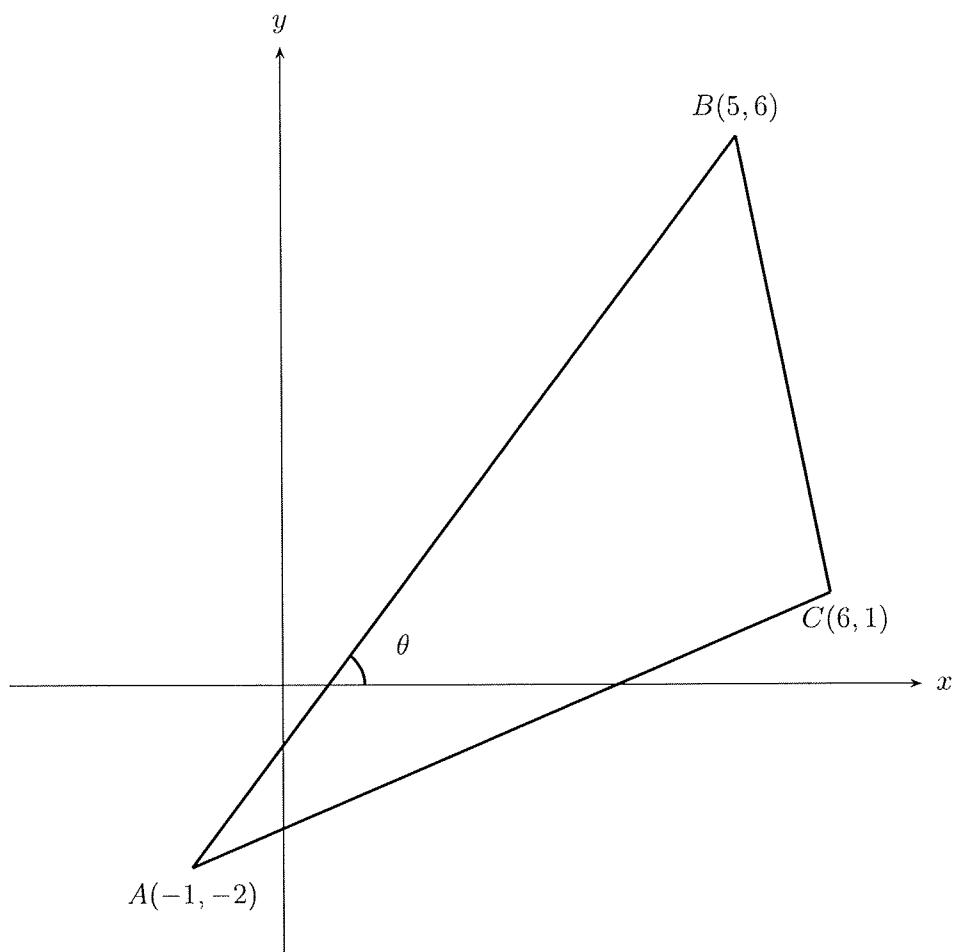
i. Find the value of x , correct to 2 decimal place. **2**

ii. Find the area of the triangle. **2**

Question 2 (8 Marks)

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Marks



- | | | |
|-----|--|---|
| (a) | Find the gradient of AB | 1 |
| (b) | Find the size of θ to the nearest degree. | 1 |
| (c) | Show that the equation of AB is $4x - 3y - 2 = 0$ | 2 |
| (d) | Find the perpendicular distance of AB from the point C . | 2 |
| (e) | Hence, find the area of the triangle ABC . | 2 |

Question 3 (9 Marks)

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Marks

- | | | |
|-----|---|---|
| (a) | Solve the following equations: | |
| | i. $\cot x = -\sqrt{3}$ for $0^\circ \leq x \leq 360^\circ$ | 2 |
| | ii. $\sqrt{2} \sin 2x = 1$ for $-180^\circ \leq x \leq 180^\circ$ | 3 |
| (b) | Prove the following identity: | 4 |

$$\frac{\cos \theta}{1 + \sin \theta} + \tan \theta = \sec \theta$$

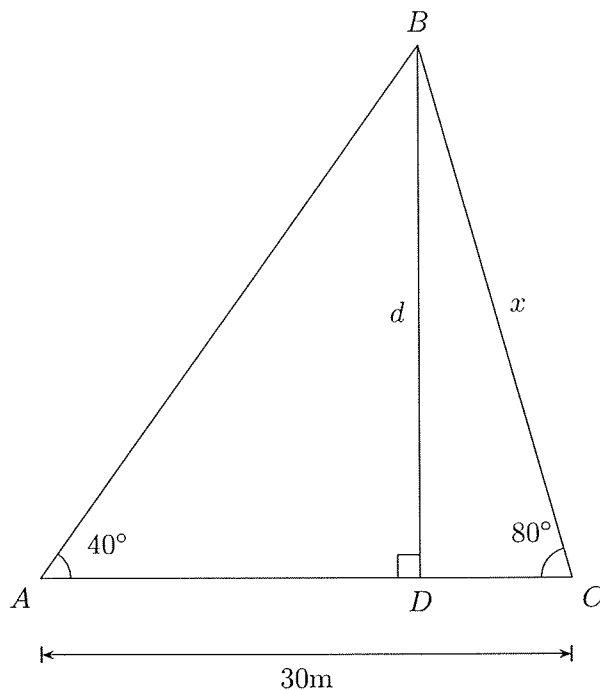
Question 4 (14 Marks)

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Marks

- (a) Solve $2 \cos^2 \theta - 5 \cos \theta = 2$, for $0 \leq \theta \leq 360^\circ$. 4

(b)



In the above triangle. $\angle ABC = 40^\circ$ and $\angle ACB = 70^\circ$. The base side, BC is 30m in length.

- | | | |
|-----|---|---|
| | i. Find the length of x . | 2 |
| | ii. Hence, find its perpendicular height h . | 1 |
| (c) | i. Sketch the curve $y = 3 \cos 2x$, for $0^\circ \leq x \leq 360^\circ$ | 2 |
| | ii. On the same diagram sketch the curve $y = 1 - \sin x$ | 3 |
| | iii. Hence find the number of solutions to the equation. | 2 |

$$1 - \sin x - 3 \cos 2x = 0, \text{ for } 0^\circ \leq x \leq 360^\circ$$

Explain your reasoning. (You are **not** required to solve the equation.)

End of paper.

Question 1

a) $\alpha = 46^\circ 3'$

b) i) $-\frac{\sqrt{3}}{2}$

ii) $\tan 60 = \sqrt{3}$

$$\therefore \frac{2(\sqrt{3})}{1 - (\sqrt{3})^2}$$

$$= \frac{2\sqrt{3}}{-2}$$

$$= -\sqrt{3}$$

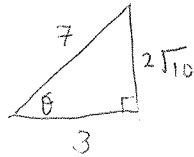
c) $\cos 55^\circ = \frac{10}{x}$

$$x = \frac{10}{\cos 55^\circ}$$

$$\approx 17.4 \text{ cm}$$

d) i) $\sec \theta = -\frac{7}{3}$

ii)



$$\therefore \sin \theta = -\frac{2\sqrt{10}}{7}$$

e) i) $x^2 = 2^2 + 4^2 - 2 \times 2 \times 4 \cos 30^\circ$

$$\hat{=} 6.14$$

$$x \hat{=} 2.48 \text{ (to 2.d.p.)}$$

ii) $\text{Area} = \frac{1}{2} \times 2 \times 4 \times \sin 30^\circ$

$$= 4 \times \frac{1}{2}$$

$$= 2 \text{ units}^2$$

Question 2

$$a) \quad m_{AB} = \frac{6 - (-2)}{5 - (-1)}$$
$$= \frac{4}{3} \quad [1]$$

$$b) \quad \tan \theta = \frac{4}{3} \quad \therefore \theta = 53^\circ \text{ (to nearest degree)} \quad [1]$$

c) using the point gradient formula

$$y - 6 = \frac{4}{3}(x - 5) \quad [1]$$

$$3y - 18 = 4x - 20$$

$$4x - 3y - 2 = 0$$

[1] for correct step to get into general form.

$$d) \quad a = 4 \quad b = -3 \quad c = -2 \quad x_1 = 6 \quad x_2 = 1$$

$$\therefore d = \frac{|4 \times 6 + (-3) \times 1 + (-2)|}{\sqrt{4^2 + (-3)^2}}$$

$$= \frac{|19|}{\sqrt{25}}$$

$$= \frac{19}{5} \text{ units.}$$

$$e) \quad AB = \sqrt{(5 - (-1))^2 + (6 - (-2))^2}$$
$$= \sqrt{36 + 64}$$
$$= \sqrt{100}$$
$$= 10$$

$$\therefore \text{Area of triangle } ABC = \frac{1}{2} \times 10 \times \frac{19}{5}$$
$$= 19 \text{ units}^2$$

Question 1

a) $\sin \alpha = 0.72 \quad \therefore \alpha = 46^\circ 3'$ [1]

b) i) $\cos 210^\circ = \cos (180^\circ + 30^\circ)$

$$= -\cos 30^\circ$$
 [1]

$$= -\frac{\sqrt{3}}{2}$$
 [1]

ii) $\tan 60^\circ = \sqrt{3}$ [1]

$$\therefore \frac{2 + \tan 60^\circ}{1 - \tan^2 60^\circ} = \frac{2 + \sqrt{3}}{1 - (\sqrt{3})^2}$$

$$= \frac{2 + \sqrt{3}}{-2}$$

$$= -\sqrt{3}$$
 [1]

c) $\cos 55^\circ = \frac{10}{x}$ [1]

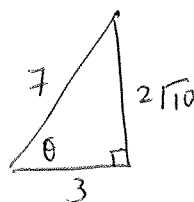
$$\therefore x = \frac{10}{\cos 55^\circ}$$

$$= 17.4 \text{ cm}$$
 [1]

d) i) $\sec \theta = \frac{1}{\cos \theta}$ [1]

$$= -\frac{7}{3}$$

(ii)



$$\therefore \sin \theta = -\frac{2\sqrt{10}}{7}$$

[1] for correct triangle

[1] for $\sin \theta$ ratio

[1] for sign

e) i) $x^2 = 2^2 + 4^2 - 2 \times 2 \times 4 \times \cos 30^\circ$ [1]

$$= 20 - 16 \cos 30^\circ$$

$$\doteq 6.14 \text{ (to 2 d.p.)}$$

$$\therefore x = \sqrt{6.14}, \text{ as } x > 0$$

$$\doteq 2.48 \text{ (to 2 d.p.)}$$
 [1]

ii) $\text{Area} = \frac{1}{2} \times 2 \times 4 \times \sin 30^\circ$ [1]

$$= 4 \times \frac{1}{2}$$

$$= 2 \text{ units}^2$$
 [1]

Question 3

a) (i) $\cot x = -\sqrt{3}$

$$\tan x = -\frac{1}{\sqrt{3}} \quad [1]$$

Since $\tan x < 0$, $90^\circ < x < 180^\circ$, $270^\circ < x < 360^\circ$

$$x = 150^\circ, 330^\circ \quad [1]$$

(ii) $\sin 2x = \frac{1}{\sqrt{2}}$ for $-180^\circ \leq x \leq 180^\circ$ [1] for change in domain
 $\therefore -360^\circ \leq 2x \leq 360^\circ$

$$2x = -315^\circ, -225^\circ, 45^\circ, 135^\circ \quad [1]$$

$$x = -\frac{315^\circ}{2}, -\frac{225^\circ}{2}, \frac{45^\circ}{2}, \frac{135^\circ}{2} \quad [1]$$

b) RTP: $\frac{\cos \theta}{1 + \sin \theta} + \tan \theta = \sec \theta$

$$\text{LHS} = \frac{\cos \theta}{1 + \sin \theta} + \tan \theta$$

$$= \frac{\cos \theta}{1 + \sin \theta} + \frac{\sin \theta}{\cos \theta}$$

$$= \frac{\cos^2 \theta + \sin \theta + \sin^2 \theta}{(1 + \sin \theta) \cos \theta}$$

$$= \frac{1 + \sin \theta}{(1 + \sin \theta) \cos \theta}$$

$$= \frac{1}{\cos \theta}$$

$$= \sec \theta$$

$$= \text{RHS}$$

Question 4

a) $2\cos^2\theta - 5\cos\theta - 2 = 0$ [1] for understanding it is quadratic in disguise.

$$\cos\theta = \frac{5 \pm \sqrt{41}}{4}$$

$$= \frac{5 + \sqrt{41}}{4} \quad \text{or} \quad \frac{5 - \sqrt{41}}{4} \quad [1]$$

$\therefore$ no solution for $\cos\theta = \frac{5 + \sqrt{41}}{4}$ for $\cos\theta = \frac{5 - \sqrt{41}}{4}$
as $\cos\theta > 1$ [1] $\theta = 110^\circ 32' \text{ or } 249^\circ 28'$ [1]

b) i) $\angle ABC = 60^\circ$ (angle sum of triangle) [1]

$$\therefore \frac{x}{\sin 40^\circ} = \frac{30}{\sin 60^\circ} \quad (\text{using the sine rule}) \quad [1]$$

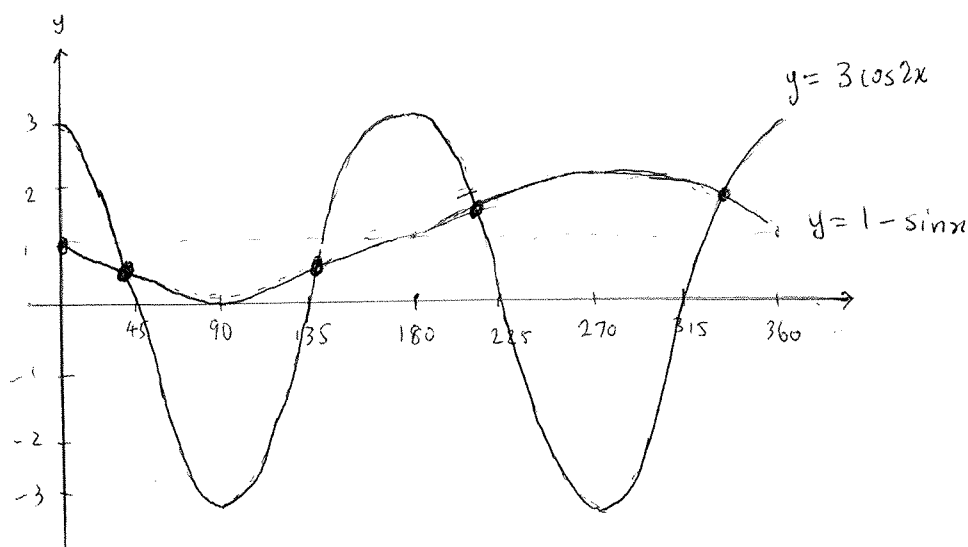
$$\therefore x = 22.3 \text{ m}$$

ii) $\sin 80^\circ = \frac{d}{22.3}$

$$\therefore d = 22.3 \times \sin 80^\circ$$

$$= 21.9 \text{ m}$$

c)



i) [1] for amplitude

[1] for period and correct shape

ii) [1] for flip of $\sin x$

[1] for translation

[1] for correct sine shape.

$$1 - \sin x - 3\cos 2x = 0$$

$$1 - \sin x = 3\cos 2x$$

$\therefore$ the solutions are the x -coordinate of the intersections of the two curves [1] reasoning

$\therefore$ there are 4 intersection points

$\therefore$ 4 solutions [1]